

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer with direct Perplexity Sonar API production experience (MS CS, 2.5+ years). Built structured prompt transformation layers over Sonar API output that improved contextual alignment ~30% and first-pass image success rate ~25-30%. Specializes in RAG integration, semantic retrieval, REST API deployment, and end-to-end AI product delivery. Strong Python, FastAPI, and JSON/REST payload handling across distributed AI systems. 5+ years of technical project ownership; comfortable with both developer teams and non-technical stakeholders.

CORE COMPETENCIES

Perplexity Sonar API Integration • RAG & Semantic Retrieval • REST API Design & Debugging • LLM Integration & Prompt Engineering • FastAPI / Docker / Python • LangChain / LlamaIndex / Vector Search • Customer-Facing Technical Delivery • Enterprise API Production Deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting

- Built **RAG systems** for clinical knowledge retrieval; improved precision by ~35%, reduced irrelevant retrieval by >30%; grounding alignment exceeded 90% in evaluation.
- Developed agentic LLM workflows with structured reasoning and grounding rules; improved stability by ~25%, reduced hallucinations by >30%.
- Packaged ML/LLM inference as **FastAPI RESTful services** with Docker, structured logging, and load simulation; reduced post-deployment defects by ~30%.
- Client-facing technical communication across healthcare enterprise stakeholders; system-limitation docs, evaluation audit trails.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

- RAG document Q&A for non-technical staff; reduced hallucinations ~30%, raised accuracy >85%. Flask-based API.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- SQL/T-SQL optimization (~20% latency reduction), Jenkins CI/CD deployment pipelines (>30% error reduction).

KEY PROJECTS

Dream Decoder - Perplexity Sonar API Integration

FastAPI + React/TypeScript/Vite/Tailwind application using **Perplexity Sonar API** for semantic dream interpretation + Replicate image synthesis. Designed **intermediate structured prompt transformation layers** between Sonar output and image generation - improved contextual alignment ~30% and first-pass image success rate ~25-30% over naive direct-prompt orchestration. Direct production experience with Sonar API payload handling, latency optimization, and auth flow management.

Agentic Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with schema-validated JSON contracts between agents, RAG-grounded CPT/ICD validation (vector-indexed), ANN-based duplicate detection, and explainable risk scoring with audit-ready reasoning traces. Demonstrates end-to-end AI system ownership including integration debugging, data quality issues, and production hardening.

Manga Lens - Multi-Provider AI Inference Chrome Extension

Production extension integrating Claude Sonnet, GPT-4o, GPT-4.1 Nano, and Ollama - REST/JSON payload debugging across multiple AI providers, auth flow management, and cross-provider reliability engineering at scale (29 sites).

EDUCATION

SKILLS

Languages: Python, JavaScript/TypeScript, SQL

LLM & APIs: Perplexity Sonar, OpenAI, Anthropic Claude, Ollama; RAG, LangChain, LlamaIndex, vector search, prompt engineering

Backend: FastAPI, Flask, REST APIs, Docker, JSON payload handling, auth flows

ML: PyTorch, scikit-learn, XGBoost, HuggingFace

Work Authorization: F-1 OPT (US-authorized now; future sponsorship required)